EXPERIMENT 12

SECURITY FEATURES AND GOVERNANCE

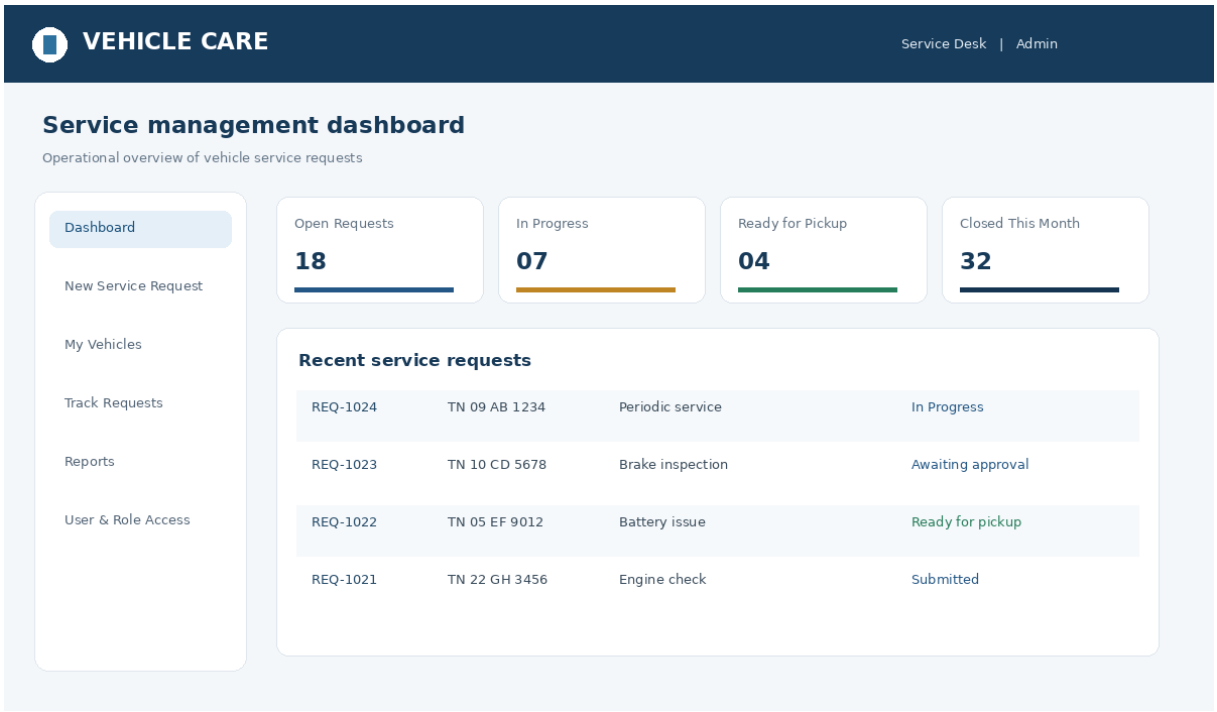
Vehicle Service Request Management System

A practical plan for protecting vehicle, customer, service, and request data in a Zoho Creator application

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PRACTICAL STUDY AND IMPLEMENTATION PLAN



Illustrative application dashboard for the vehicle service workflow

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1. Aim and scope

To outline security, governance, and administration practices for a Vehicle Service Request Management System built in Zoho Creator. The plan covers account protection, role-based access, data permissions, safe Deluge scripting, managed connections, controlled releases, and audit review. Customer contact details, vehicle registration information, service notes, and invoices must be handled with appropriate access restrictions.

2. Procedure

Step	Planned activity
1. Identify sensitive data	Review customer details, vehicle registration numbers, service descriptions, invoices, and appointment records.
2. Protect accounts	Use strong passwords and multi-factor authentication; consider single sign-on where available.
3. Assign roles	Define Administrator, Service Advisor, Technician, and Customer roles using least privilege.
4. Set access rules	Apply form, report, record, and field permissions based on role and request ownership.
5. Secure code and connections	Validate submitted fields and store integration credentials in managed connections, not in scripts.
6. Govern releases	Separate development and testing, review changes, approve releases, and monitor audit logs.

3. Output — layered security

Layer	Purpose
Account and sign-in	Strong passwords, multi-factor sign-in, and session controls reduce unauthorised access.
Application access	Profiles and roles determine which forms, reports, and actions each user can open.
Data access	Record and field criteria restrict users to the vehicles and service requests relevant to them.
Logic and integrations	Validate inputs, restrict connections, and avoid embedding secrets in Deluge scripts.
Governance and audit	Use approvals, release notes, audit trails, backups, and periodic permission reviews.

These safeguards form defence in depth: if one control is missed, other controls can still reduce exposure.

3.1 Proposed permission matrix

The matrix is a proposed design. Verify actual profile settings in the live Creator application before deployment.

Area	Administrator	Service Advisor	Technician	Customer
Customer / vehicle records	Full control	Create / view / update	Assigned jobs only	Own records
Service requests	All records	Create / assign / update	Assigned requests	Own requests
Service notes	All notes	View / manage	Add job notes	Limited status
Invoices	All invoices	Create / view	No access	Own invoices
Users / connections	Manage	No access	No access	No access

3.2 Access-control decision flow

Check	Pass	Fail
Authentication	Confirm the signed-in user's identity.	Deny access and request sign-in.
Role permissions	Check requested form, report, and action permissions.	Deny the action when permission is missing.
Record criteria	Show only the vehicle and service records permitted for that user.	Restrict visibility or reject the action.
Audit	Record important changes for review.	Investigate unusual or repeated failures.

3.3 Secure scripting and connections

Validate required fields before accepting a service request. Registration number, request category, preferred date, and issue description should be checked for missing or invalid values. Reject dates in the past where inappropriate, and restrict status changes to authorised roles. Use a managed connection for calendar, notification, or other external integrations; never hard-code passwords, tokens, or API secrets in Deluge scripts. Test code and permissions before release.

3.4 Change lifecycle

Develop	Test	Approve	Publish	Monitor
Build forms, workflows, and role rules in development.	Test valid and invalid requests in a test environment.	Review functionality, data access, and release notes.	Publish the approved version to production.	Review audit trails, errors, and user feedback.

3.5 Risks and controls

Risk	Example	Control
Unauthorised access	A former employee can still view customer records.	Remove access promptly and review sign-in safeguards.
Excess privilege	A technician can edit billing or user settings.	Apply least privilege and separate role permissions.
Data leakage	A report exposes customer phone numbers or invoices.	Review sharing and export settings; avoid public links.
Insecure scripts	Missing validation or credentials embedded in code.	Validate input, use managed connections, review scripts.
Uncontrolled release	A live workflow update interrupts service booking.	Test changes and approve them before publication.
Poor traceability	A request status changes without an accountable record.	Review audit logs and record important status updates.

3.6 Administration checklist

Area	Routine task	Frequency
Users	Assign the correct profile; remove leavers promptly.	On change
Permissions	Review access to requests, invoices, and exports.	Quarterly
Sign-in	Review multi-factor authentication and account security.	Quarterly
Connections	Limit integrations to required users and applications.	Quarterly
Changes	Test workflows before every production release.	Every release
Audit and recovery	Review unusual activity and confirm backup readiness.	Monthly / as planned

4. Application screenshots

The following interface images are illustrative mock-ups prepared for this experiment to show how the security and governance plan maps to the vehicle service workflow. They are not captured from a live Zoho Creator deployment.

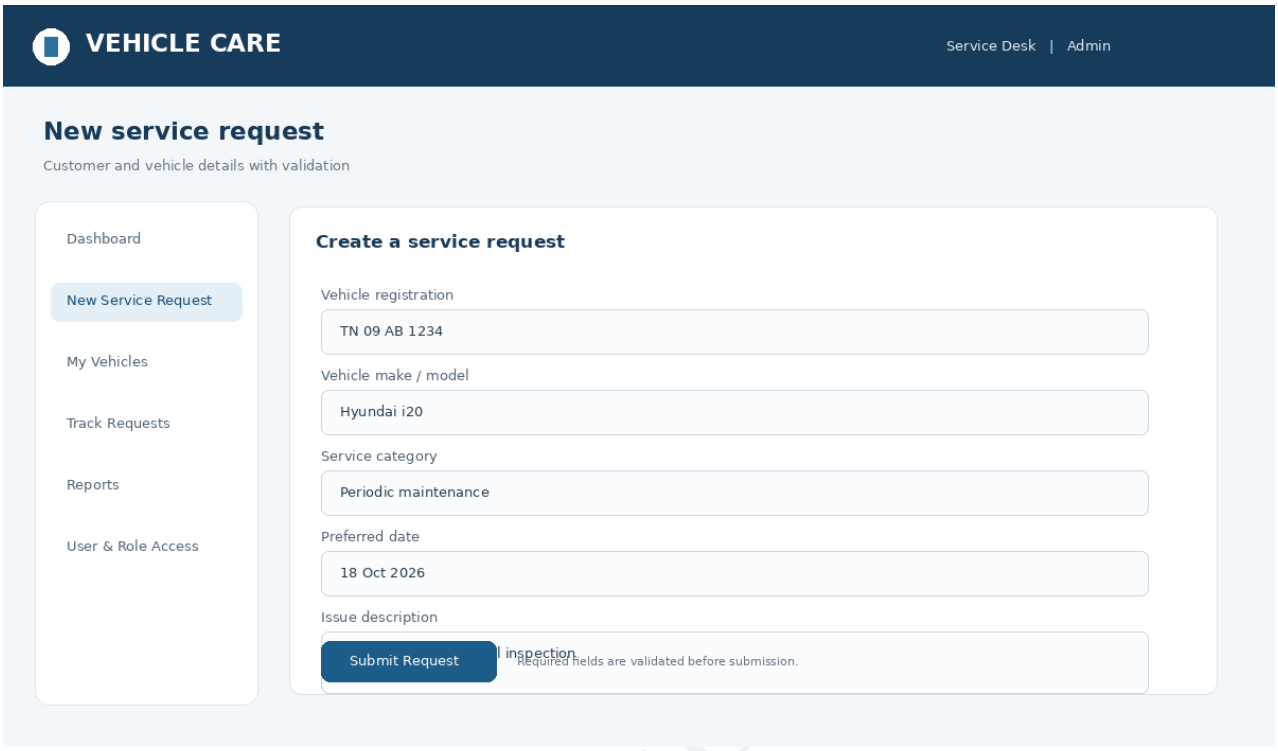


Figure 1. New service request form with input validation.

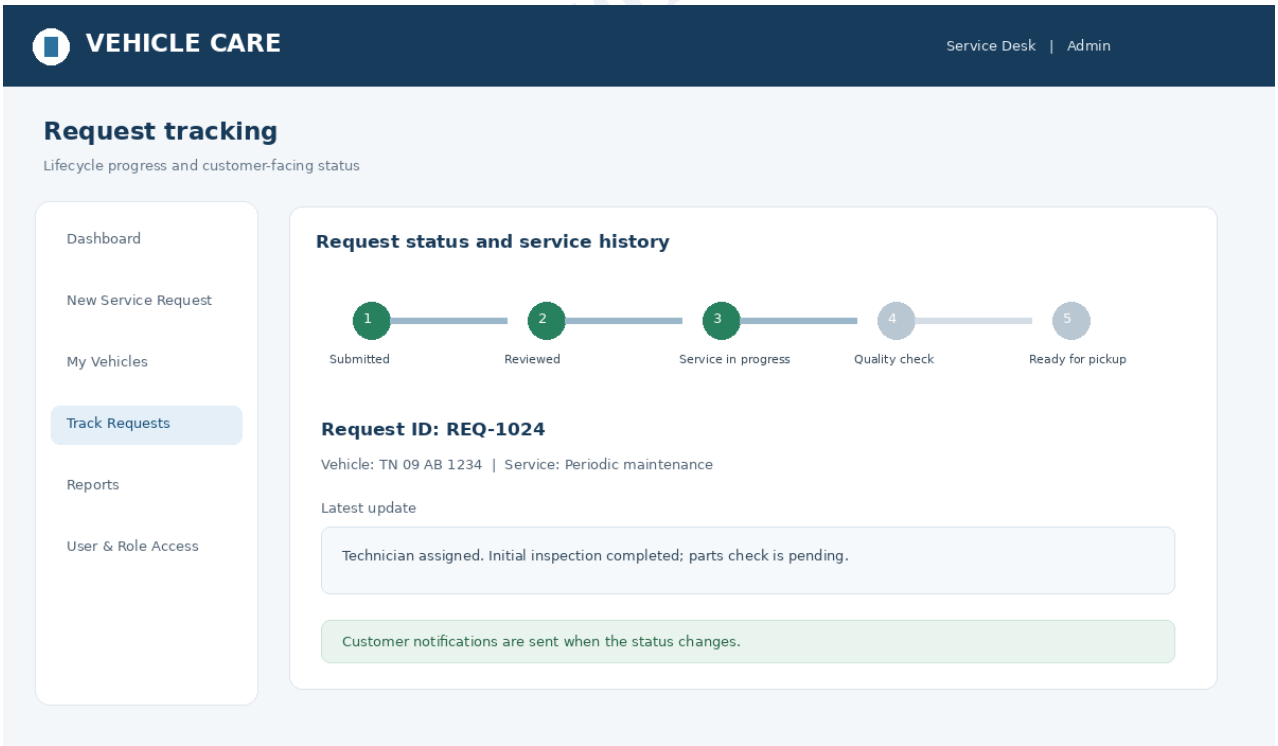


Figure 2. Request tracking screen showing status progression and updates.

5. Administration view and final result

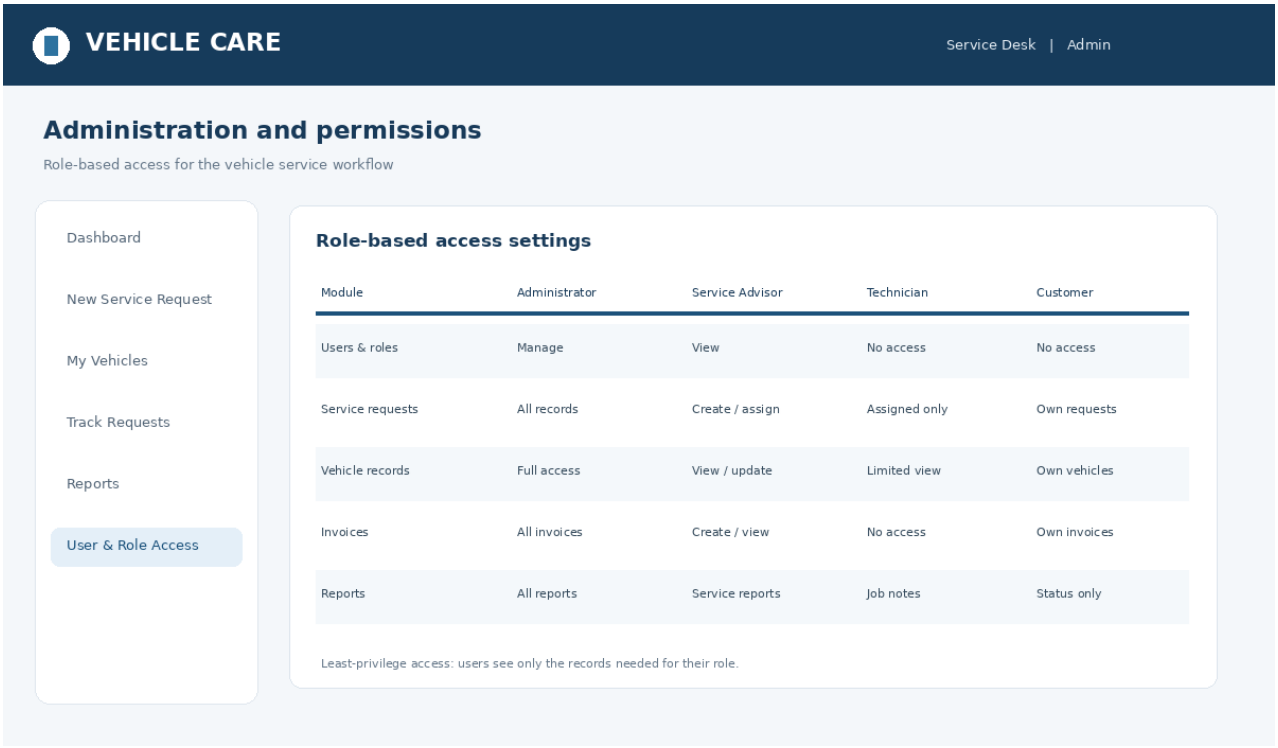


Figure 3. Proposed role-based access matrix for the service management application.

Final result

A proposed security and governance plan was prepared for the Vehicle Service Request Management System. It includes account protection, role-based permissions, restricted access to vehicle and service records, safe scripting, managed integrations, controlled releases, audit review, and routine administration checks. The screenshots are illustrative design views; the configuration and security controls must be tested and verified in the live application before the system is used with real customer data.

Precaution: Do not use real customer details in demonstration screenshots. Keep production data private and grant access only to authorised users.

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